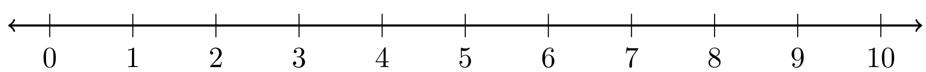
# Approximating Irrational Roots on a Number Line

*Trapping a square root between two whole numbers, estimating it to the nearest tenth, plotting it on a number line, and ordering roots with rational numbers*

**Directions.**

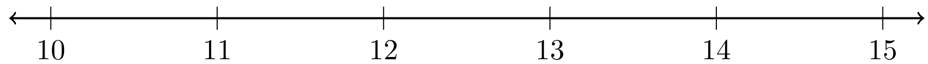
- A square root that is not a perfect square is *irrational*: its decimal never ends and never repeats, so every answer here is an **approximation**, written with $\approx$.
- Trap the root first: find the perfect square just below the radicand and the perfect square just above it. The root lies between their roots.
- To get the tenth, test tenths by squaring: if $6.8^2$ is too small and $6.9^2$ is too big, the root is between them — take whichever tenth is closer.
- Mark each answer on the number line printed with the problem, and write the numerical answer on the answer blank.
- No calculators.

**1.** Between which two consecutive whole numbers does $\sqrt{47}$ lie? Write your answer as an inequality, and mark the approximate position of $\sqrt{47}$ on the number line with a dot labelled $A$.



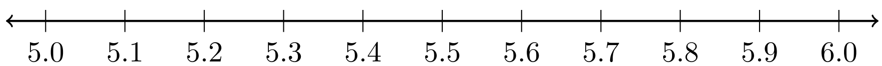

Answer: ________________

**2.** Between which two consecutive whole numbers does $\sqrt{150}$ lie? Write your answer as an inequality, and mark the approximate position of $\sqrt{150}$ on the number line with a dot labelled $B$.



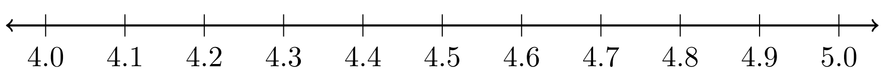

Answer: ________________

**3.** Estimate $\sqrt{30}$ to the nearest tenth. Show the two tenths you squared to decide, then plot your estimate on the number line.




Answer: ________________

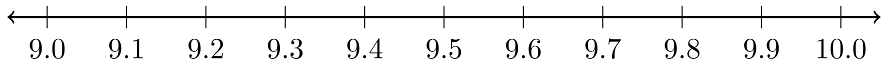
**4.** Plot $\sqrt{20}$ and 4.5 on the number line below, then write $<$ or $>$ in the blank so the statement is true.




$\sqrt{20}$ ______ 4.5

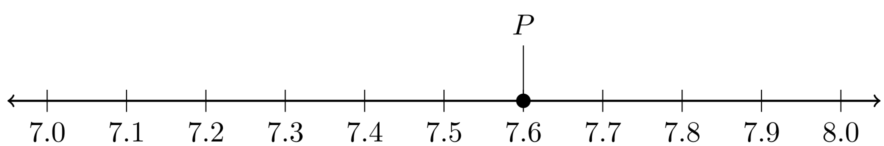
Answer: ________________

**5.** Estimate $\sqrt{83}$ to the nearest tenth and plot it on the number line. Show the squaring you used to choose between the two tenths.



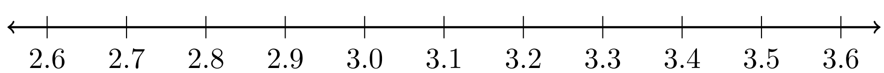

Answer: ________________

**6.** The point $P$ is already marked on the number line below. Exactly one of $\sqrt{55}$, $\sqrt{58}$ and $\sqrt{62}$ belongs at $P$. Which one is it? Write the root, and give its value to the nearest tenth.



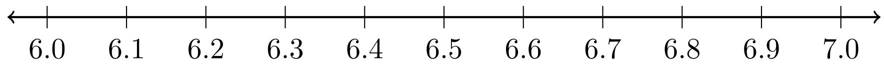

Answer: ________________

**7.** Plot 2.7, $\sqrt{8}$, 3.1 and $\sqrt{10}$ on the number line, then list all four in order from least to greatest.



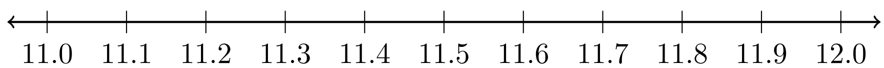

Answer: ________________

**8.** Plot $\sqrt{41}$ and $\sqrt{45}$ on the number line below, and write each one to the nearest tenth.



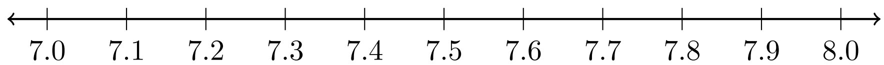

Answer: ________________

**9.** A square patio has an area of 132 square feet. Its side length in feet is $\sqrt{132}$. Estimate that side length to the nearest tenth of a foot and plot it on the number line.




**Answer:** ________________ ft

**10.** Nadia says that $\sqrt{50} \approx 7.0$, because the nearest perfect square root is 7 and she rounded to it. Plot $\sqrt{50}$ on the number line, give its value to the nearest tenth, and explain in one sentence why Nadia's rounding rule gives the wrong tenth.




Answer: __________________